
Intermediate Activation Sampling Is Not Output Sampling

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Abstract

Neural networks usually expose stochasticity at the final prediction, where a label or token is sampled from a softmax distribution. We ask what changes when sampling is moved into an intermediate representation instead. We compare deterministic inference, final-softmax sampling, categorical hidden activation sampling, stochastic activation pruning, Gaussian hidden noise, and Monte Carlo dropout in one controlled CNN pipeline on MNIST and FASHION-MNIST. Across three seeds, stochastic methods are evaluated with $S \in \{1, 5, 20\}$ Monte Carlo samples and measured with accuracy, negative log likelihood, expected calibration error, predictive entropy, and sample variance. The main result is clear: hidden activation sampling is not equivalent to output sampling. A naive categorical sampler over penultimate hidden dimensions reduces accuracy by 19.13 percentage points on MNIST and 22.46 points on FASHION-MNIST at $S = 20$, while increasing ECE from 0.002 to 0.254 on MNIST and from 0.010 to 0.202 on FASHION-MNIST. In contrast, stochastic activation pruning and Gaussian hidden noise preserve accuracy within 0.04 points on MNIST and within 0.15 points on FASHION-MNIST, while mostly increasing entropy and predictive variance. Final-softmax sampling preserves accuracy at $S = 20$ but worsens empirical sampled-probability NLL. These results show that intermediate stochasticity is a representation-level intervention, not output sampling moved earlier, and that useful hidden sampling likely requires a better distribution and sample-aware training.

1 Introduction

Sampling from a final softmax is a familiar operation: the network computes logits, converts them to probabilities, and draws a class or token from that distribution. The hidden layers that produced those logits are usually treated differently. They are deterministic vectors that the classifier head reads once. Our main question is simple: what happens if an intermediate activation vector is also treated as a distribution and sampled from?

This question matters because the location of stochasticity changes what uncertainty can mean. Output sampling changes the readout while leaving the computation intact. Hidden-layer sampling changes the representation before later layers act on it. That distinction is important for calibration, prediction diversity, robustness claims, and any method that tries to express uncertainty without placing a full distribution over weights.

Prior work already shows that stochastic hidden computations can be useful. Stochastic activation pruning samples activation indices with probabilities tied to activation magnitude Dhillon et al. [2018]. Stochastic neural networks can model Gaussian activation uncertainty Yu et al. [2021], and activation-level uncertainty can be represented with stochastic activation functions Morales-Alvarez et al. [2021]. Dropout, variational dropout, information bottlenecks, and stochastic depth also inject randomness before the final prediction Gal and Ghahramani [2016], Kingma et al. [2015], Achille and Soatto [2016], Alemi et al. [2017], Huang et al. [2016]. However, these lines of work usually

optimize specialized objectives or target robustness, compression, or uncertainty estimation. They rarely isolate the narrower comparison: hidden activation sampling versus final-softmax sampling under the same trained model and evaluation pipeline.

We examine that comparison directly. figure 1 summarizes the setup. We train compact CNN classifiers with a 128-dimensional penultimate hidden layer on MNIST and FASHION-MNIST, then evaluate deterministic inference, final-softmax label sampling, categorical sampling over hidden dimensions, SAP-style hidden masking, Gaussian hidden perturbations, and MC-DROPOUT. Each stochastic method is evaluated with $S \in \{1, 5, 20\}$ samples, and we average predictive probabilities across samples.

The result is not subtle. At $S = 20$, HIDDEN-CATEGORICAL reduces accuracy from 98.55% to 79.41% on MNIST and from 88.77% to 66.30% on FASHION-MNIST. It also raises predictive entropy by 1.265 on MNIST and 1.178 on FASHION-MNIST. By contrast, SAP and GAUSSIAN hidden sampling preserve deterministic accuracy almost exactly, while final-softmax sampling preserves accuracy but worsens empirical sampled-probability NLL from 0.044 to 0.112 on MNIST and from 0.310 to 0.734 on FASHION-MNIST.

Our contributions are:

- We define a controlled comparison between output-only sampling and intermediate activation sampling in the same trained CNN pipeline.
- We evaluate categorical hidden sampling, SAP-style sampling, Gaussian activation noise, and MC-DROPOUT across two datasets, three seeds, and three Monte Carlo sample counts.
- We show that naive categorical hidden sampling is highly disruptive, while SAP and Gaussian hidden noise preserve accuracy and mainly change uncertainty statistics.
- We identify sample-aware training and structured hidden distributions as the next step for turning hidden sampling into a useful modeling tool.

The rest of the paper reviews related work in stochastic hidden representations (section 2), describes the experimental methodology (section 3), reports the quantitative results (section 4), and discusses limitations and future directions (section 5).

2 Related Work

Activation sampling and pruning. The closest prior method is stochastic activation pruning (SAP), which samples activation indices with probability proportional to activation magnitude and rescales survivors to preserve the expected activation Dhillon et al. [2018]. SAP was proposed as a post-hoc stochastic defense, but its mechanism is directly relevant to our question because it turns hidden activations into a sampling distribution. More recent p-bit activation work also studies internal stochastic activations and sample-aware inference Ghantasala et al. [2026]. Unlike these papers, we do not propose a new defense or hardware mechanism. We use SAP and categorical hidden sampling as controlled interventions to ask whether hidden sampling behaves like output sampling.

Activation uncertainty and activation noise. Several methods put uncertainty at the activation level rather than only at the weights. Yu et al. predict Gaussian activation means and variances, then sample hidden activations during forward passes Yu et al. [2021]. Morales-Alvarez et al. move uncertainty into activation functions with Gaussian-process activation models Morales-Alvarez et al. [2021]. Noisy activation functions inject noise into nonlinearities to improve optimization and exploration Gulcehre et al. [2016]. The local reparameterization trick also converts weight uncertainty into local activation noise to reduce gradient variance Kingma et al. [2015]. We build on this view that hidden activations can be random variables, but we focus on post-hoc inference behavior under matched deterministic models.

Dropout and stochastic bottlenecks. Dropout is a standard hidden-stochastic baseline. Its Bayesian interpretation motivates Monte Carlo dropout at test time as an approximate uncertainty estimator Gal and Ghahramani [2016]. Information Dropout and the variational information bottleneck use noisy intermediate representations to control information flow Achille and Soatto [2016], Alemi et al. [2017]. Stochastic depth randomly drops residual blocks during training Huang et al. [2016]. These methods differ from our hidden categorical sampler because their stochasticity is either trained into the model, applied multiplicatively, or structured at the layer level. We include MC-DROPOUT to anchor our results against a familiar stochastic baseline.

Input image	Feature extractor	Penultimate vector	Sampling intervention	Prediction
x	$g_\theta(x)$	$h \in \mathbb{R}^{128}$	$\tilde{h}^{(s)} \sim \{\text{HIDDEN-CATEGORICAL, SAP, GAUSSIAN, MC-DROPOUT}\}$	$\frac{1}{S} \sum_s \text{softmax}(c_\theta(\tilde{h}^{(s)}))$
x	$g_\theta(x)$	h	$y^{(s)} \sim \text{Categorical}(\text{softmax}(c_\theta(h)))$	output-sample counts

Figure 1: Experimental comparison. Hidden sampling modifies the penultimate representation before the classifier head. Output sampling leaves the representation fixed and samples only from the final softmax.

Discrete stochastic nodes. Sampling hidden dimensions creates discrete stochastic computation. Gradient estimators for stochastic neurons and conditional computation provide one route for training such models Bengio et al. [2013]. Gumbel-Softmax and Concrete relaxations provide differentiable approximations to categorical sampling Jang et al. [2017], Maddison et al. [2017]. Our experiments do not train through categorical hidden samples; they test the post-hoc effect first. The strong degradation we observe suggests that differentiable or sample-aware training is likely necessary if categorical hidden sampling is to become a practical method.

Stochastic robustness evaluation. Stochastic classifiers require careful robustness evaluation because attack and inference sample counts can change conclusions Daubener and Fischer [2022]. We do not run robustness attacks in this first study. This choice keeps the experiment focused on clean accuracy, calibration, entropy, and variance, and avoids overclaiming robustness from stochasticity alone.

3 Methodology

Problem setup. Let f_θ be a classifier with feature extractor g_θ , penultimate representation $h \in \mathbb{R}^d$, and linear classifier head c_θ . For an input x , deterministic inference computes

$$h = g_\theta(x), \quad p = \text{softmax}(c_\theta(h)). \quad (1)$$

The deterministic prediction is $\arg \max_y p_y$. Hidden sampling replaces h with a stochastic $\tilde{h}^{(s)}$ before the classifier head, then averages predictive probabilities:

$$\bar{p}_S(x) = \frac{1}{S} \sum_{s=1}^S \text{softmax}(c_\theta(\tilde{h}^{(s)})). \quad (2)$$

This differs from final-softmax sampling, which leaves h fixed, draws labels $y^{(s)} \sim \text{Categorical}(p)$, and estimates an empirical predictive distribution from sample counts.

Inference methods. We evaluate six inference conditions. DETERMINISTIC uses the standard forward pass. OUTPUT-SAMPLE samples labels from the final softmax and averages one-hot outcomes across S samples. HIDDEN-CATEGORICAL forms $q_i = \text{softmax}(|h_i|/\tau)$ over the 128 penultimate dimensions, draws a target 25% of dimensions, and rescales sampled coordinates by inverse probability. SAP samples a Bernoulli mask over hidden dimensions with probabilities proportional to $|h_i|$, targets a 35% keep fraction, and rescales survivors by inverse probability Dhillon et al. [2018]. GAUSSIAN adds zero-mean noise proportional to the per-example hidden standard deviation, with scale 0.35. MC-DROPOUT uses a dropout-trained model with dropout active at inference; DROPOUT-EVAL uses the same dropout-trained model with dropout disabled.

Datasets. We use local HuggingFace snapshots of MNIST LeCun et al. [1998] and FASHION-MNIST Xiao et al. [2017]. For each seed, the original training split is stratified into 20,000 training examples and 5,000 validation examples, and the official 10,000-example test set is held out for final evaluation. Normalization uses only the training subset mean and standard deviation. table 1 summarizes the splits.

Model and training. The classifier is a compact CNN with two convolutional blocks and a 128-dimensional penultimate hidden layer. We train two variants for each dataset and seed: a deterministic model with dropout probability 0, and a dropout model with dropout probability 0.25 before the classifier head. We use seeds 42, 123, and 456. Training uses batch size 256, Adam-style optimization with learning rate 0.001, weight decay 0.0001, up to 8 epochs, early stopping patience 3, and mixed precision on ‘cuda:0’. The recorded environment used Python 3.12.8, PyTorch 2.12.0+cu130, CUDA 13.0, and an NVIDIA RTX A6000 GPU.

Table 1: Dataset construction. Training and validation splits are stratified subsets of the original training split; test sets are the official test sets. Normalization uses only the training subset for each seed.

Dataset	Train	Validation	Test	Notes
MNIST	20,000	5,000	10,000	Stratified training subset; official test set
FASHION-MNIST	20,000	5,000	10,000	Balanced stratified subset; official test set

Table 2: Primary test metrics. Accuracy is mean $\pm$ standard deviation across three seeds. Stochastic methods use $S = 20$. Bold marks the best accuracy and lowest NLL/ECE within each dataset.

Dataset	Model	Method	Acc. (%)	NLL	ECE	Entropy	Prob. var.
FASHION-MNIST	deterministic	DETERMINISTIC	88.77 ± 0.16	0.310	0.010	0.286	0.0000
FASHION-MNIST	deterministic	GAUSSIAN	88.75 ± 0.16	0.310	0.008	0.317	0.0013
FASHION-MNIST	deterministic	HIDDEN-CATEGORICAL	66.30 ± 13.40	1.032	0.202	1.464	0.0228
FASHION-MNIST	deterministic	OUTPUT-SAMPLE	88.53 ± 0.13	0.734	0.015	0.249	0.0143
FASHION-MNIST	deterministic	SAP	88.61 ± 0.03	0.320	0.024	0.371	0.0037
FASHION-MNIST	dropout	DROPOUT-EVAL	89.06 ± 0.36	0.305	0.008	0.282	0.0000
FASHION-MNIST	dropout	MC-DROPOUT	89.00 ± 0.25	0.307	0.009	0.328	0.0011
MNIST	deterministic	DETERMINISTIC	98.55 ± 0.12	0.044	0.002	0.040	0.0000
MNIST	deterministic	GAUSSIAN	98.53 ± 0.12	0.044	0.003	0.053	0.0004
MNIST	deterministic	HIDDEN-CATEGORICAL	79.41 ± 9.27	0.833	0.254	1.304	0.0197
MNIST	deterministic	OUTPUT-SAMPLE	98.51 ± 0.18	0.112	0.003	0.032	0.0019
MNIST	deterministic	SAP	98.51 ± 0.12	0.055	0.017	0.100	0.0017
MNIST	dropout	DROPOUT-EVAL	98.81 ± 0.13	0.036	0.002	0.035	0.0000
MNIST	dropout	MC-DROPOUT	98.82 ± 0.14	0.038	0.006	0.059	0.0005

Metrics and statistics. We report accuracy, NLL, ECE with 10 bins Guo et al. [2017], BRIER score, predictive entropy, mean probability variance across Monte Carlo samples, per-class accuracy, and evaluation time. Stochastic methods use $S \in \{1, 5, 20\}$ samples. Primary tables report $S = 20$ unless noted. For deterministic-model stochastic methods, we compare each $S = 20$ condition against deterministic inference using paired seed-level differences, paired t tests, Wilcoxon signed-rank tests where supported, Cohen’s d_z , 95% confidence intervals, and Holm correction within dataset/metric families. With only three seeds, we treat p -values as descriptive and emphasize effect sizes and raw differences.

4 Results

Main result. table 2 reports primary metrics. The central pattern is that hidden categorical sampling behaves unlike output sampling. On MNIST, HIDDEN-CATEGORICAL at $S = 20$ reaches 79.41% accuracy, compared with 98.55% for deterministic inference and 98.51% for OUTPUT-SAMPLE. On FASHION-MNIST, it reaches 66.30%, compared with 88.77% and 88.53%. It also produces much higher NLL, ECE, entropy, and probability variance.

SAP and Gaussian sampling preserve accuracy. The less discrete hidden interventions are much less disruptive. On MNIST, SAP and GAUSSIAN both remain within 0.04 percentage points of deterministic accuracy at $S = 20$. On FASHION-MNIST, SAP is 0.15 points below deterministic accuracy and GAUSSIAN is 0.02 points below. Their main visible effect is uncertainty: SAP increases entropy from 0.040 to 0.100 on MNIST and from 0.286 to 0.371 on FASHION-MNIST, while GAUSSIAN increases entropy to 0.053 and 0.317.

Output sampling is a readout-level intervention. Final-softmax sampling preserves top-line accuracy when S is large enough because the underlying softmax probabilities are unchanged. However, the evaluated sampled distribution is a count distribution over 20 one-hot draws. This coarse empirical distribution worsens NLL from 0.044 to 0.112 on MNIST and from 0.310 to 0.734 on FASHION-MNIST, even though accuracy remains within 0.23 points of deterministic inference.

Monte Carlo averaging helps but does not fix categorical hidden sampling. table 3 shows the sample-count sweep. More samples improve every stochastic method, but the categorical hidden sampler remains far below deterministic inference. On MNIST, it rises from 62.28% at $S = 1$ to 79.41% at $S = 20$; on FASHION-MNIST, it rises from 50.06% to 66.30%. In contrast, SAP and

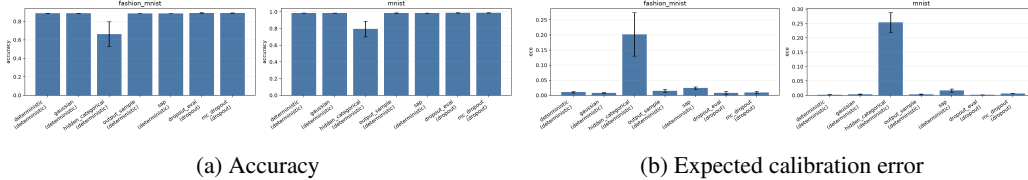


Figure 2: Primary accuracy and calibration results. HIDDEN-CATEGORICAL sharply reduces accuracy and increases ECE on both datasets. SAP, GAUSSIAN, and MC-DROPOUT preserve accuracy much more closely.

Table 3: Accuracy across Monte Carlo sample counts. Values are mean $\pm$ standard deviation across three seeds. Increasing S helps stochastic methods, but HIDDEN-CATEGORICAL remains far below deterministic inference.

Dataset	Model	Method	$S = 1$	$S = 5$	$S = 20$
MNIST	deterministic	OUTPUT-SAMPLE	97.86 ± 0.19	98.39 ± 0.19	98.51 ± 0.18
MNIST	deterministic	HIDDEN-CATEGORICAL	62.28 ± 6.26	74.36 ± 8.36	79.41 ± 9.27
MNIST	deterministic	SAP	97.22 ± 0.23	98.35 ± 0.07	98.51 ± 0.12
MNIST	deterministic	GAUSSIAN	98.20 ± 0.18	98.49 ± 0.11	98.53 ± 0.12
MNIST	dropout	MC-DROPOUT	98.44 ± 0.17	98.77 ± 0.18	98.82 ± 0.14
FASHION-MNIST	deterministic	OUTPUT-SAMPLE	84.29 ± 0.40	87.58 ± 0.37	88.53 ± 0.13
FASHION-MNIST	deterministic	HIDDEN-CATEGORICAL	50.06 ± 7.19	60.52 ± 11.45	66.30 ± 13.40
FASHION-MNIST	deterministic	SAP	86.04 ± 0.38	88.29 ± 0.20	88.61 ± 0.03
FASHION-MNIST	deterministic	GAUSSIAN	87.89 ± 0.14	88.66 ± 0.07	88.75 ± 0.16
FASHION-MNIST	dropout	MC-DROPOUT	88.23 ± 0.17	88.82 ± 0.17	89.00 ± 0.25

GAUSSIAN approach deterministic accuracy by $S = 20$, and MC-DROPOUT slightly improves the dropout-trained model’s averaged prediction.

Paired statistical comparisons. table 4 summarizes key paired differences against deterministic inference for deterministic-model methods at $S = 20$. HIDDEN-CATEGORICAL lowers accuracy by 19.13 percentage points on MNIST and 22.46 points on FASHION-MNIST. These accuracy differences have large paired effect sizes ($d_z = -2.07$ and -1.66), although Holm-corrected p -values are not below 0.05 with only three seeds. Calibration and entropy shifts are more consistent: HIDDEN-CATEGORICAL increases ECE by 0.252 on MNIST ($p_{\text{Holm}} = 0.026$) and entropy by 1.265 on MNIST ($p_{\text{Holm}} = 0.015$) and 1.178 on FASHION-MNIST ($p_{\text{Holm}} = 0.011$).

5 Discussion

Hidden sampling changes the computation. The experiments support the behavioral hypothesis: sampling intermediate activations is not equivalent to sampling from the final softmax. OUTPUT-SAMPLE leaves the learned representation and classifier head intact, so averaging enough samples recovers the decision behavior of the original model. Hidden sampling changes the vector consumed by the classifier head, so it can change class margins, confidence, calibration, and errors before the final softmax is ever formed.

The categorical sampler is too destructive post hoc. The strongest result is also a warning. A direct “softmax over hidden magnitudes, sample dimensions, and rescale” rule removes too much representational information from a normally trained CNN. Even expectation-style rescaling and 20 Monte Carlo samples do not restore deterministic accuracy. This does not mean categorical hidden sampling is impossible; it means the naive post-hoc version is mismatched to the representation. More plausible designs include sample-aware training, grouped or channel-level sampling, Gumbel-Softmax training relaxations, or a learned stochastic bottleneck.

Continuous and magnitude-aware hidden noise is less disruptive. SAP and GAUSSIAN behave differently because they preserve more of the penultimate representation. SAP keeps high-magnitude coordinates more often and rescales survivors, while GAUSSIAN perturbs the hidden vector continuously. Both interventions preserve accuracy on MNIST and FASHION-MNIST while

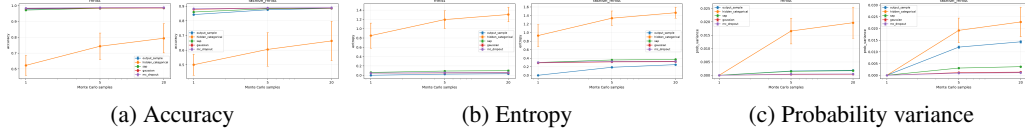


Figure 3: Sample-count effects for stochastic inference. Monte Carlo averaging stabilizes predictions, but HIDDEN-CATEGORICAL remains high-entropy and low-accuracy even at $S = 20$.

Table 4: Key paired differences versus deterministic inference for deterministic-model methods at $S = 20$. Accuracy differences are percentage points; other differences are raw metric units. Holm-corrected p -values are descriptive because $n = 3$ seeds.

Dataset	Method	Δ Acc.	Δ NLL	Δ ECE	Δ Entropy
MNIST	GAUSSIAN	-0.01	0.001	0.001	0.013
MNIST	HIDDEN-CATEGORICAL	-19.13	0.789	0.252	1.265
MNIST	OUTPUT-SAMPLE	-0.03	0.069	0.001	-0.007
MNIST	SAP	-0.03	0.011	0.015	0.060
FASHION-MNIST	GAUSSIAN	-0.02	0.001	-0.002	0.031
FASHION-MNIST	HIDDEN-CATEGORICAL	-22.46	0.723	0.192	1.178
FASHION-MNIST	OUTPUT-SAMPLE	-0.23	0.424	0.004	-0.037
FASHION-MNIST	SAP	-0.15	0.010	0.014	0.085

increasing entropy and probability variance. In this setup, their value is not better accuracy; it is a controlled way to change uncertainty statistics without destroying task performance.

Interpreting output-sampling NLL. The OUTPUT-SAMPLE NLL numbers should be read as empirical sampled-distribution quality, not the model’s analytic softmax quality. With $S = 20$, the sampled probability assigned to a class is a multiple of $1/20$, so the distribution is coarse. This explains why accuracy can remain high while empirical NLL worsens. The contrast is still useful because it shows that output sampling and hidden sampling fail in different ways.

Limitations. This study uses only MNIST and FASHION-MNIST, a compact CNN, three seeds, and a single penultimate sampling location. Larger datasets, residual networks, transformers, spatial feature maps, and channel-wise sampling may behave differently. We also evaluate post-hoc sampling rather than training with the stochastic mechanism active. Finally, we do not make robustness claims. Stochastic defenses require expectation-over-transformation attacks and careful control of attack-time sample counts Daubener and Fischer [2022].

Broader implications. The results argue for treating intermediate stochasticity as a modeling choice rather than a cosmetic change to inference. Useful hidden sampling needs a distribution that respects the geometry of learned representations and an objective that teaches downstream layers to consume sampled states. Without that alignment, hidden sampling can produce uncertainty, but the uncertainty comes from damaging the representation rather than from calibrated ambiguity.

6 Conclusion

We tested whether sampling an intermediate activation distribution changes neural-network behavior differently from deterministic inference and final-softmax sampling. In a controlled CNN study on MNIST and FASHION-MNIST, the answer is yes. Naive categorical hidden sampling sharply reduces accuracy and worsens calibration, while SAP-style and Gaussian hidden sampling preserve accuracy and mainly increase entropy and predictive variance. Final-softmax sampling behaves differently again: it preserves accuracy with enough samples, but its empirical sampled probabilities degrade NLL.

The key takeaway is that hidden activation sampling is a representation-level intervention. It cannot be assumed to inherit the benign behavior of output sampling. The next experiments should train with the sampling mechanism active, test structured channel or spatial sampling, and move to CIFAR-scale models with robustness and calibration evaluations.

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